

Dynamic VR Modulation with EEG Feedback for Psychedelic Simulation

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ABSTRACT

This study explores approaches to combining real-time electroencephalographic (EEG) with VR to generate synthetic psychedelic experiences. Using a passive brain-computer interface (BCI) that dynamically adjusts virtual content, we aim to elicit neurocognitive states similar to those that characterize psychedelic trips - including attenuation of self-referential thought and diminished engagement of the default mode network. Preliminary results indicate that this type of personalized VR experience can modulate cortical activities and elicit unique states of altered consciousness. By mimicking hallucinatory experiences, we aim in the future to replicate the cognitive and emotional benefits of psychedelic agents without the associated risks.

Index Terms: Virtual Reality (VR), Digital Psychedelics, EEG, Brain-Computer Interface (BCI), Default Mode Network (DMN), Therapeutic Interventions, Human-Computer Interaction (HCI).

1 INTRODUCTION

Psychedelics and virtual reality (VR) share significant phenomenological and therapeutic similarities, both offering immersive experiences that can alter perception, evoke awe, and promote self-transcendence [1]. Both methods have shown promise in reducing anxiety and depression (reviewed in [1]). However, traditional psychedelics necessitate careful monitoring in therapeutic settings since they carry risks, such as psychosis, anxiety, and physical health concerns. Therefore, some studies have begun exploring "digital psychedelics," using VR to mimic psychedelic effects without side effects. Examples include Isness-D [4], Deep Dream-stylized videos [11], which enhanced cognitive flexibility [10], and Psyrrreal, a VR experience mimicking common psychedelic-induced hallucinations, which led to lasting decreases in depressive symptoms [5].

We propose that even greater benefits could be obtained by customizing simulated psychedelic experiences to elicit mental and brain states that approximate the feeling of ego dissolution and suppression of default mode network (DMN) activities that are hallmarks of authentic psychedelic experience. The DMN, which is less active during external engagement, plays a role in self-referential processing and higher cognitive functions [8]. Psychedelics are known to disrupt brain activity, increasing signal diversity and reducing oscillatory power within the DMN network [3, 9]. Our approach employs a closed-loop brain-computer interface (BCI) system that leverages EEG to adaptively regulate an immersive virtual psychedelic experience. By dynamically adjusting the virtual environment based on real-time EEG feedback, we ensure that scenes most effective in reducing oscillatory EEG activities associated with DMN activity are displayed for extended durations. This tailoring of stimuli to each participant's unique neural

responses aims to mirror the effects of psychedelics, and to facilitate altered states of consciousness. These states are subsequently evaluated through post-hoc questionnaires.

2 GRAPHICAL DISPLAY AND VIRTUAL REALITY (VR)

Traditional VR hallucination simulations lack adaptive control. Thus, we propose a BCI-compatible experience using the Deep Dream algorithm to create various hallucination levels. By preprocessing and switching scenes in real-time based on EEG features, we combine Deep Dream with Gaussian Splatting [6] to create immersive 3D psychedelic environments. Additionally, we developed a real-time style transfer system and a Sphere Tracing renderer in Unity for fractal visualizations based on EEG activity.

To create psychedelic fractal scenes, we employ constructive solid geometry primitives and operators, applying transformations to form iterated function system fractals. These models, rendered with sphere tracing, are animated with various folds and rotations. Using the Deep Dream and a style transfer network, it is possible to progressively enhance the psychedelic effect.



Figure 1: The VR scenes featured original lab space renderings (top left), altered visuals using Deep Dream (top center), style transfer (top right), and dynamic kaleidoscopic 3D fractals (lower panels).

3 EEG AND BRAIN COMPUTER INTERFACE (BCI)

We recorded EEG data using a research-grade 64-channel EEG system (BioSemi, Amsterdam). The data was processed online using the NeuroPype platform (Intheon, CA), applying a bandpass filter (1-40 Hz), rereferencing to the common average, and using ICA for artifact correction. We continuously computed 5-second segments using a moving window technique, and transformed each into the frequency domain using an FFT. Based on literature, we focused on 10 posterior EEG channels related to parietal alpha activities and DMN engagement [7]. The alpha power (8-12Hz) of these 10 channels was averaged, and the resulting continuous stream of posterior alpha power was imported into Unity.

Participants viewed a default landscape in VR for 2 minutes to establish an eyes-open resting baseline. Posterior alpha power during this baseline served as a threshold for comparing continuous alpha power during the digital psychedelics experience. When alpha power exceeded the baseline, it triggered a scene transition. In this way, scenes that effectively suppressed alpha power were viewed longer, while those that did not were switched more quickly. Each scene was displayed for max. 60 seconds, with the entire VR experience lasting 4 to 8 minutes.

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4 USER STUDY

4.1 Participants and Procedure

Six healthy adults participated (3 female; 3 male) (age range: 19–47, $M=26.3$). All were neurologically healthy. Participants visited the lab on two separate days. During the experimental session, we first recorded the 2-minute resting state EEG baseline, followed by the Digital Psychedelics experience (Figure 1). After the VR experience, participants completed the 5D-Altered States of Consciousness (ASC)[2] Scale, which is a validated tool for assessing changes in consciousness relative to normal waking states. During the control session, we recorded another 2-minute resting state baseline and then presented the control experience, which consisted of 360° neutral nature videos that required passive viewing, like the psychedelic experience, but were selected because they were unlikely to induce emotional or altered states. Afterwards, participants again completed the 5D-ASC.

4.2 Results

The Psychedelic experience substantially altered participants' states of consciousness compared to the control condition. The most notable changes were in experiencing a Blissful State and Altered Meanings of Percepts subscales (Figure 2). Supplementary Figure S1 illustrates the standard deviations for each condition and subscale for further insights into the variability of the data.

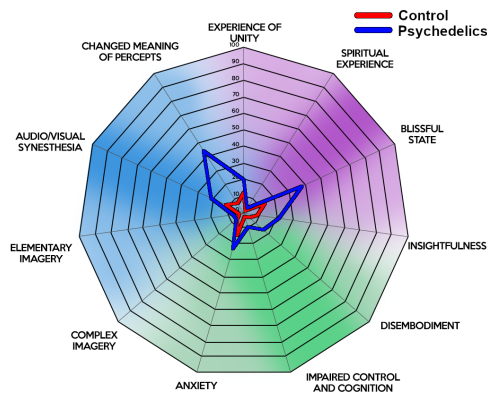


Figure 2: Post-VR experience responses to the 5-Dimensional Altered States of Consciousness Rating Scale (5D-ASC) per condition.

The EEG results indicate that posterior alpha power was lower during the BCI-guided psychedelic experience compared to the control condition (Figure 3). This outcome demonstrates that our system successfully adapted and individualized the VR experience to promote suppression of alpha band oscillatory activity. As expected, the most effective scenes for alpha power suppression varied among participants. Participants had different “best-suited scenes,” indicating that the individualized approach was crucial for optimizing the effectiveness of the VR experience in alpha power (Supplementary Figure S2).

5 DISCUSSION

We present a cutting-edge BCI system that utilizes EEG to dynamically tailor an immersive virtual psychedelic experience, serving as a personalized tool to explore altered states of consciousness.

Findings from the 5D-ASC Scale suggest that our BCI-driven VR experience induces stronger consciousness changes relative to neutral control videos along parameters that are characteristic of psychedelic phenomenology. Our approach builds upon previous research emphasizing the therapeutic potential of VR and

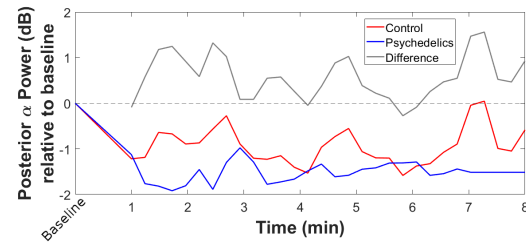


Figure 3: Posterior alpha power over time per condition.

psychedelics [1, 12], incorporating real-time EEG feedback to customize VR experiences and boost their efficacy. The observed suppression of posterior alpha power during the psychedelic experience suggests that our EEG-guided system can induce targeted neurodynamic states. Currently, only scene switching is controlled by EEG, but work is ongoing to modulate additional scene parameters with a broader range of EEG features in order to induce brain states that more closely approximate a psychedelic trip.

Integrating VR with BCI technology guided by EEG metrics has significant potential for innovative therapeutic applications. By leveraging the immersive qualities of VR and real-time neural feedback, we can craft personalized experiences aimed at enhancing mental well-being and cognitive functioning. This approach opens up exciting possibilities for developing effective VR therapies that can enhance cognitive flexibility and reduce depressive symptoms in a manner comparable to that of psychedelic treatments, but without the associated risks and side effects.

REFERENCES

- [1] J. S. Aday, C. C. Davoli, and E. K. Bloesch. Psychedelics and virtual reality: parallels and applications. *Therapeutic Advances in Psychopharmacology*, 10:2045125320948356, 2020. 1, 2
- [2] A. Dittrich, D. Lamparter, and M. Maurer. 5d-asc: Questionnaire for the assessment of altered states of consciousness. *A short introduction. Zurich, Switzerland: PSIN PLUS.*, 2010. 2
- [3] N. Farnes, B. E. Juel, A. S. Nilsen, L. G. Romundstad, and J. F. Storm. Increased signal diversity/complexity of spontaneous eeg, but not evoked eeg responses, in ketamine-induced psychedelic state in humans. *Plos one*, 15(11):e0242056, 2020. 1
- [4] D. R. Glowacki, R. R. Williams, M. D. Wonnacott, O. M. Maynard, R. Freire, J. E. Pike, and M. Chatziapostolou. Group vr experiences can produce ego attenuation and connectedness comparable to psychedelics. *Scientific Reports*, 12(1):8995, 2022. 1
- [5] K. K. Kaup, M. Vasser, K. Tulver, M. Munk, J. Pikamäe, and J. Aru. Psychedelic replications in virtual reality and their potential as a therapeutic instrument: an open-label feasibility study. *Frontiers in Psychiatry*, 14:1088896, 2023. 1
- [6] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023. 1
- [7] G. Knyazev, J. Slobodskoj-Plusnin, A. Bocharov, and L. Pyrkova. The default mode network and eeg alpha oscillations: an independent component analysis. *Brain research*, 1402:67–79, 2011. 1
- [8] V. Menon. 20 years of the default mode network: A review and synthesis. *Neuron*, 111(16):2469–2487, 2023. 1
- [9] P. D. Petridis. A psychedelic state arises from desynchronized brain activity, 2024. 1
- [10] C. Rastelli, A. Greco, Y. N. Kenett, C. Finocchiaro, and N. De Pisapia. Simulated visual hallucinations in virtual reality enhance cognitive flexibility. *Scientific reports*, 12(1):4027, 2022. 1
- [11] K. Suzuki, W. Roseboom, D. J. Schwartzman, and A. K. Seth. A deep-dream virtual reality platform for studying altered perceptual phenomenology. *Scientific reports*, 7(1):1–11, 2017. 1
- [12] S. Woolfe. Virtual reality simulates psychedelic experiences. 2023. 2